

Software Testing  
Course's Code: CSE 453  
Automated Tool Support for Testing  
(Automated Mock Framework )[Chapter 5]

# Chapter 5

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## **Chapter 5. Automated Tool support for Testing**

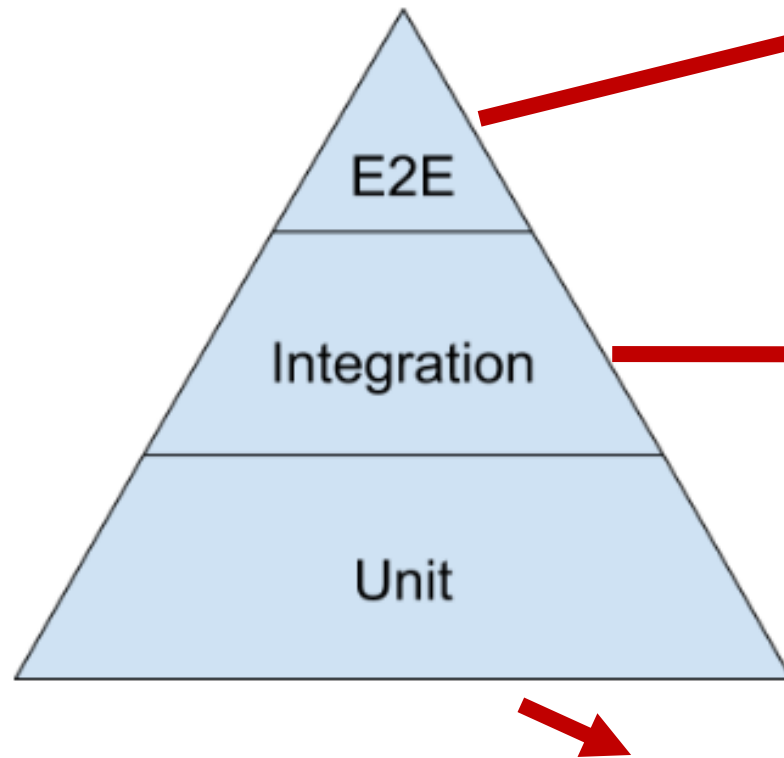
5.1 Automated Mock Framework

**5.2 API Testing**

5.3 Web Testing

5.4 Behaviour Driven Development (BDD)

# Automated Tool Support towards Different levels of testing pyramid



- Junit, TestNG, Mockito for Java language
- Jest for java script
- Nunit for .net languages
- pytest for python

- Selenium, Cypress, Playwright, Appium
- Postman for API testing tool manually,
- Rest Assured for automated API testing tool, [This also follow BDD]
- Pact for Consumer Driven Contract testing tool,
- Cucumber follows BDD [Behavior Driven Development]
- Mockito to test interfaces between different components of a software
- WireMocks for API testing

- **This is not the Complete List**
- **More tools exist**
- **Go for a search**

# API Testing

## What is an API?

- Application
- Programming
- Interface

- Let's start our journey into API testing with a basic introduction to API's, and what they are.
- First of all, what does API stand for? API stands for Application Programming Interface, which basically just means that it's one piece of code talking to another piece of code.
- This could be **your website talking to a database back end**, it could be **a mobile application talking to an authentication service**.
- There are all sorts of configurations that it could be, but it's essentially **one piece of code talking to another piece of code**.

# API Testing

## The API Contract

- ▶ Request
  - ▶ Endpoint
  - ▶ Headers
  - ▶ Body
- ▶ Response
  - ▶ Status Code
  - ▶ Headers
  - ▶ Body

- The way that an API works is through a contract.
- First of all, you have a request, which includes and endpoint, headers and a body.
- When the service receives this request, it responds with its own set of headers, body, and also a status code.

# API Testing

## APIs are for developers

- ▶ Correct
- ▶ Provides Value
- ▶ Intuitive

- Even though the customer is the person who ultimately benefits from the API, in that case, it's important to remember that the **end user for an API is actually the developer of an application.**
- The developers are the ones who are using the APIs to add functionality to their web pages, or their mobile applications, or whatever.
- Things that are important about APIs for developers are:

# API Testing

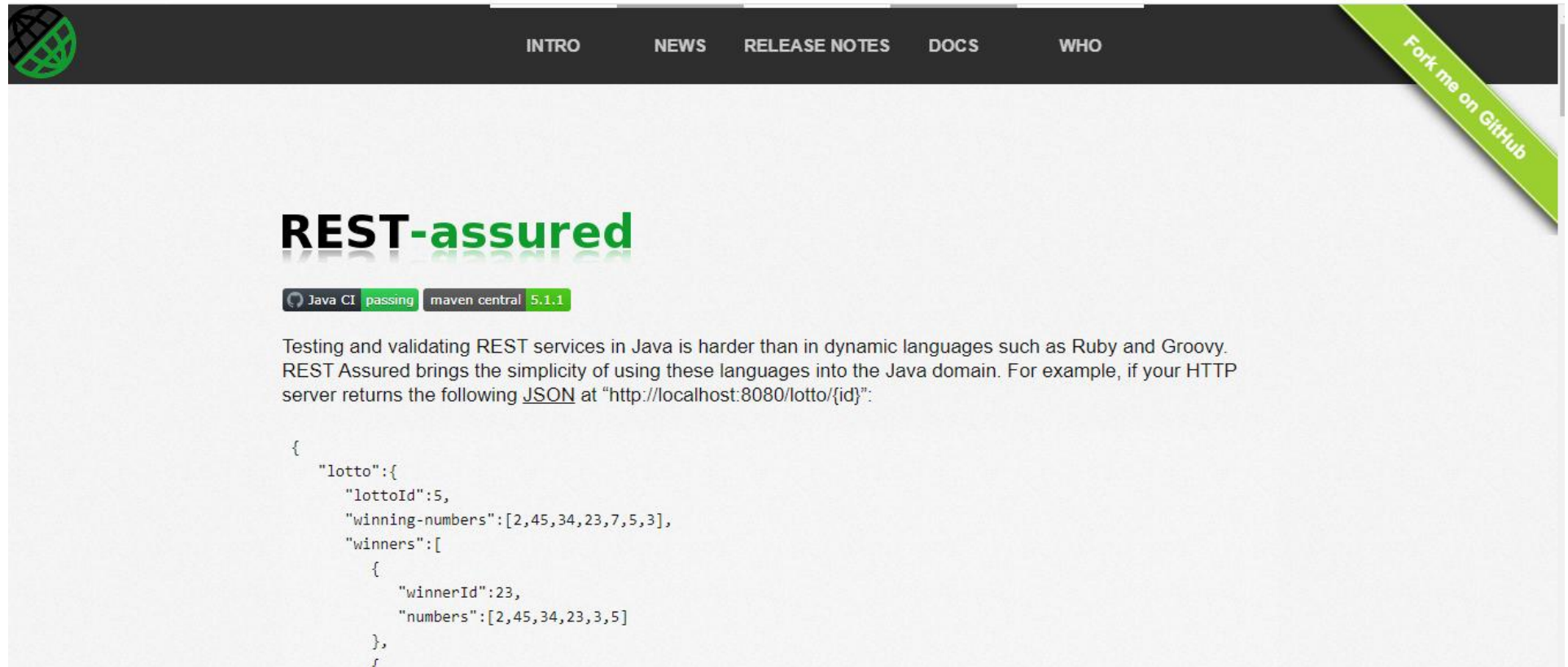
## APIs are for developers

- ▶ Correct
- ▶ Provides Value
- ▶ Intuitive

- That the **APIs are correct**, in that the contract is honored in the way that it's supposed to be.
- That they **provide value, that they're doing something that the application needs**, to show information to the customer or to process data or whatever.
- And that they're intuitive. Basically, meaning that **the API should be easy for a developer to work with**. In the same way that a good UI will be easy for a customer to navigate, a good API should also be intuitive for developers to program with.
- One thing that's really crucial to remember, is that **APIs are essentially public information**.

# RestAssured: a tool to test API

<https://rest-assured.io/>





# RestAssured: a tool to test API

- REST Assured is a Java DSL, or Domain-Specific Language, that allows you to write powerful, readable, and maintainable tests in Java for your RESTful APIs.
- Your REST Assured tests run on top of existing unit testing frameworks like JUnit or TestNG.
- Need to create a maven project and need to add the following dependencies: rest-assured, json-path, xml-path, junit-jupiter-engine
- We will test the following API: <http://zippopotam.us/>
- REST Assured provides support for all known HTTP methods. GET, POST, PUT, PATCH, DELETE, and all other existing HTTP methods.
- REST Assured also supports the Gherkin syntax — the Given/ When/Then [BDD]. However REST Assured doesn't force you to use it. It just gives you the option to do so.

# RestAssured: a tool to test API

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See an example of Different Requests in the following links

<https://devqa.io/rest-assured-api-requests-examples/>